Uniwersytet Wrocławski

Recommender Systems

Matrices, Graphs, Sequences

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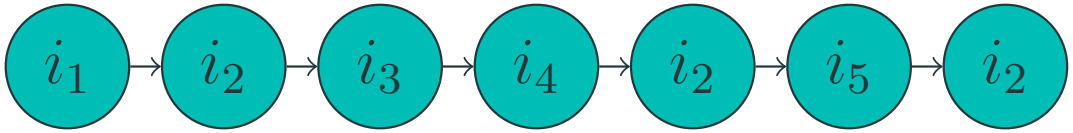
Data representations

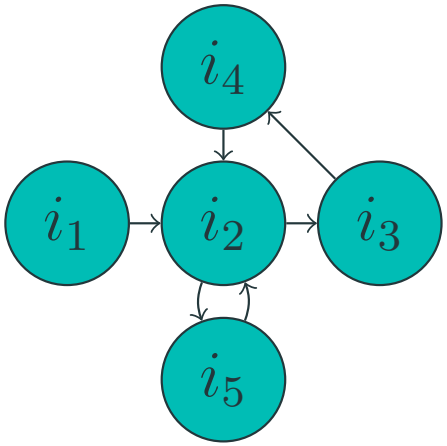
- matrix

$$\begin{bmatrix} 0 & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & 0 \\ 0 & 0 & \mathbf{1} & 0 & \mathbf{1} & \mathbf{1} & 0 & 0 & \mathbf{1} & \mathbf{1} \\ 0 & 0 & 0 & \mathbf{1} & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 \\ 0 & \mathbf{1} & 0 & 0 & 0 & 0 & \mathbf{1} & 0 & 0 & 0 \end{bmatrix}$$

- sequence



- graph



Algebraic Matrix Factorization

Pure algebra:

- SVD (on blackboard last week)
- NMF

More advanced options:

- SVD++
- FunkSVD
- ...

Loss functions for MF (continuation)

Known loss functions used for RS

- MSE (regression)
- cross-entropy (classification)

Probabilitic Matrix Factorization

- User prior: $P(\mathbf{P}) = \prod_u \mathcal{N}(\mathbf{p}_u|0, \sigma_U^2 I)$
- Item prior: $P(\mathbf{Q}) = \prod_i \mathcal{N}(\mathbf{q}_i|0, \sigma_I^2 I)$
- Rating probability: $R_{i,j} \sim \mathcal{N}(.|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)$,
- 1_{ui} - indicates if user i has interacted with item j
- Likelihood:

$$p(R|\mathbf{P}, \mathbf{Q}, \sigma^2) = \prod_{u,i} \left[\mathcal{N}(r_{ui}|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)\right]^{1_{ui}}$$

- Posterior:

$$p(R|\sigma_U^2, \sigma_V^2, \sigma^2) = \prod_{u,i} \left[\mathcal{N}(r_{ui}|\mathbf{p_u} \times \mathbf{q_i}, \sigma^2)\mathcal{N}(\mathbf{p_u}|0, \sigma_U^2 I)\mathcal{N}(\mathbf{q_i}|0, \sigma_V^2 I)\right]^{1_{ui}}$$

Ruslan Salakhutdinov and Andriy Mnih. 2007. Probabilistic Matrix Factorization. In Proceedings of the 21st International Conference on Neural Information Processing Systems (NIPS'07). Curran Associates Inc., Red Hook, NY, USA, 1257–1264.

Loss function

Maximizing the log-posterior over movie and user features with hyperparameters (i.e. the observation noise variance and prior variances) kept fixed is equivalent to minimizing the sum-of-squared-errors objective function with L_2 regularization terms:

$$E = \frac{1}{2} \sum_{u,i} 1_{ui} (r_{ui} - \mathbf{p_u} \times \mathbf{q_i})^2 + \frac{\lambda_U}{2} \sum_u \|\mathbf{p_u}\|^2 + \frac{\lambda_I}{2} \sum_i \|\mathbf{q_i}\|^2$$

where $\lambda_U = \sigma^2/\sigma_U^2$, $\lambda_V = \sigma^2/\sigma_V^2$.

Optimization: gradient descent in $\mathbf{P}$ and $\mathbf{Q}$.

Scales linearly with number of observations.

Additional Tricks

- Scale Ratings to $[0, 1]$: $t(x) = (x - 1)/(K - 1)$
- Scale Dot-products to $[0, 1]$: $g(x) = 1/(1 + \exp(-x))$

- **Constrained version**:

$$\mathbf{p_u} = \tilde{\mathbf{p}}_u + \frac{\sum_i 1_{ui} W_i}{\sum_i 1_{ui}}$$

where $\tilde{\mathbf{p}}_u$ is the user-specific part and $W \in R^{D \times M}$ is the item specific part.

Bayesian Personalized Ranking

Idea:

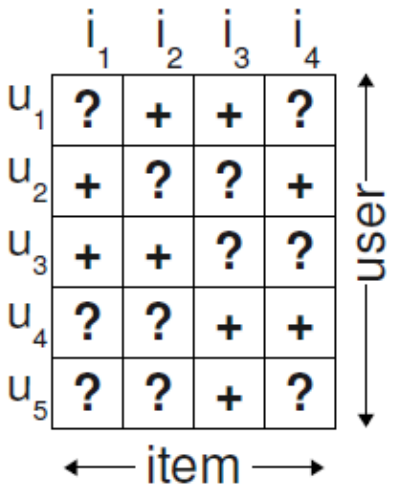
Provide a ranking of items – total order on all items:

$$\forall i, j \in I : i \neq j \Rightarrow i \geq_u j \vee j \geq_u i \quad \text{(totality)}$$
$$\forall i, j \in I : i \geq_u j \wedge j \geq_u i \Rightarrow i = j \quad \text{(antisymmetry)}$$
$$\forall i, j, k \in I : i \geq_u j \wedge j \geq_u k \Rightarrow i \geq_u k \quad \text{(transitivity)}$$

Intuition: optimize for correctly ranking item pairs instead of scoring single items

Rendle, S., Freudenthaler, C., Gantner, Z., & Schmidt-Thieme, L. (2009, June). BPR: Bayesian personalized ranking from implicit feedback.

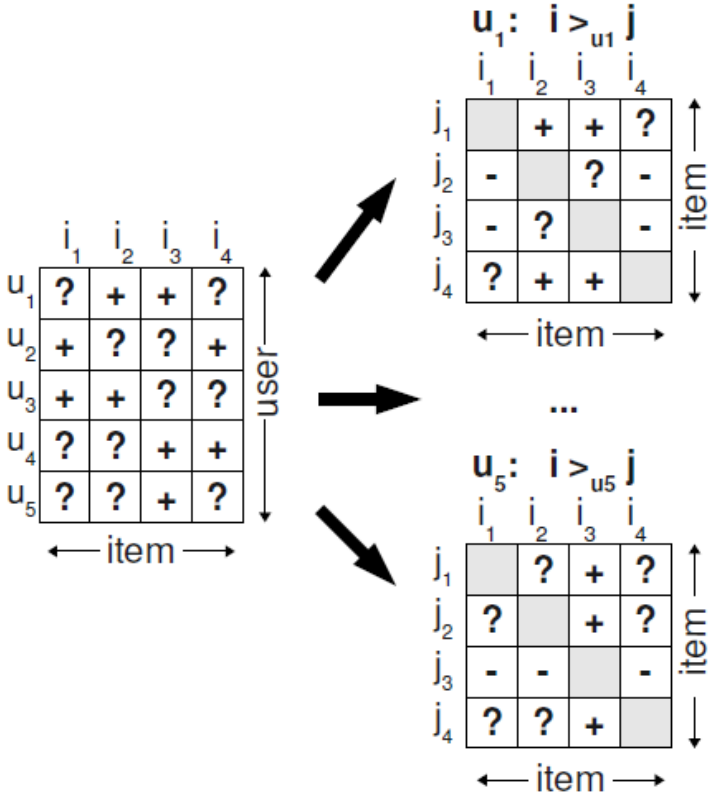
Data Representation:



$$D_S := \left\{(u, i, j) \mid i \in I_u^+ \wedge j \in I \setminus I_u^+\right\}$$

triples: $(user, positive, negative)$

Example



Loss function

$$\prod_{u \in U} p(\geq_u | \Theta) = \prod_{(ui,j) \in D_S} p(i \geq_u j | \Theta)$$

$$p(i \geq_u j | \Theta) := \sigma(\hat{x}_{ui,j}(\Theta))$$

$$\begin{aligned} BPR - OPT &:= \ln p(\Theta | \geq_u) \\ &= \dots \\ &= \sum_{(ui,j) \in D_S} \ln \sigma(\hat{x}_{ui,j}) - \lambda_\Theta \|\Theta\|^2 \end{aligned}$$

Uniformity & Alignment

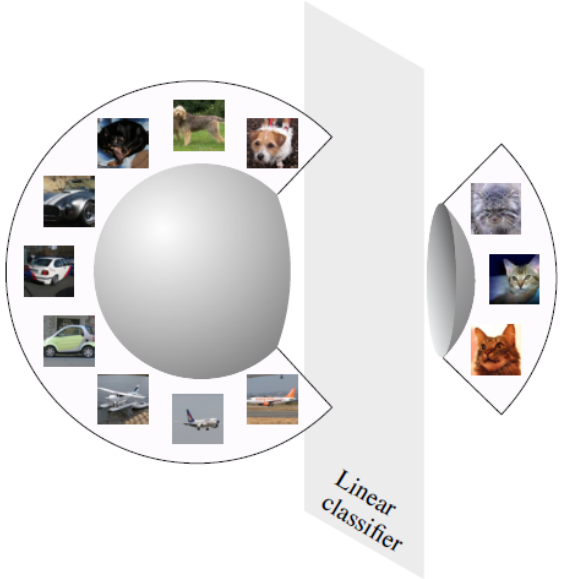


Figure 2: **Hypersphere:** When classes are well-clustered (forming spherical caps), they are linearly separable. The same does not hold for Euclidean spaces.

Tongzhou Wang and Phillip Isola. 2020. Understanding contrastive representation learning through alignment and uniformity on the hypersphere. In Proceedings of the 37th International Conference on Machine Learning (ICML'20), Vol. 119. JMLR.org, Article 921, 9929–9939.

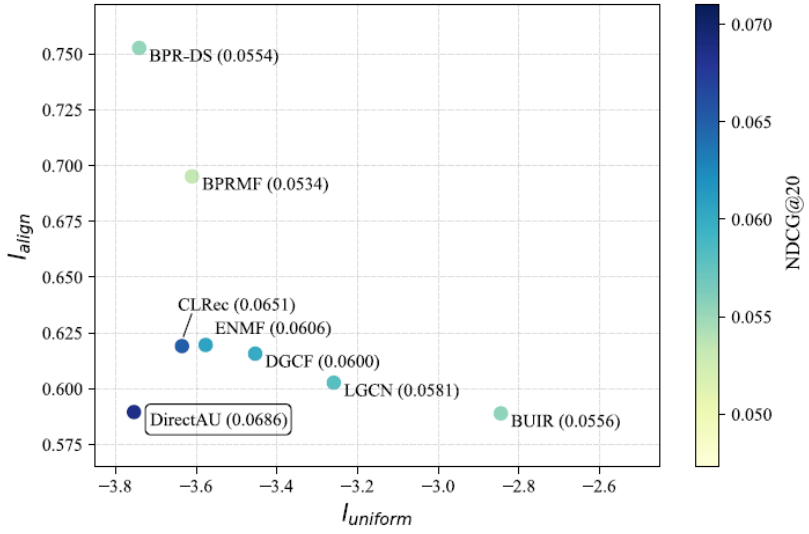
Uniformity & Alignment for RS

$$l_{align} = E_{(u,i)\sim positive} \|f(u) - f(i)\|^2$$

$$l_{uniformity} = \log E_{(u,u')} \frac{1}{2} e^{-2\|f(u)-f(u')\|^2} + \log E_{(i,i')} \frac{1}{2} e^{-2\|f(i)-f(i')\|^2}$$

Chenyang Wang, Yuanqing Yu, Weizhi Ma, Min Zhang, Chong Chen, Yiqun Liu, and Shaoping Ma. 2022. Towards Representation Alignment and Uniformity in Collaborative Filtering. In Proceedings of the 28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD '22). Association for Computing Machinery, New York, NY, USA, 1816–1825. https://doi.org/10.1145/3534678.3539253

Relation to other metrics



Source: https://arxiv.org/abs/2206.12811

Interactions as Tabular Data

Factorization Machines (tabular data alike)

Combination of SVM (with polynomial kernel) and matrix factorization (to handle sparsity).

	Feature vector x																									Target y	
	User				Movie					Other Movies rated					Time	Last Movie rated											
	A	B	C	...	TI	NH	SW	ST	...	TI	NH	SW	ST	...		TI	NH	SW	ST	...							
$x^{(1)}$	1	0	0	...	1	0	0	0	...	0.3	0.3	0.3	0	...	13	0	0	0	0	...		5	$y^{(1)}$				
$x^{(2)}$	1	0	0	...	0	1	0	0	...	0.3	0.3	0.3	0	...	14	1	0	0	0	...		3	$y^{(2)}$				
$x^{(3)}$	1	0	0	...	0	0	1	0	...	0.3	0.3	0.3	0	...	16	0	1	0	0	...		1	$y^{(2)}$				
$x^{(4)}$	0	1	0	...	0	0	1	0	...	0	0	0.5	0.5	...	5	0	0	0	0	...		4	$y^{(3)}$				
$x^{(5)}$	0	1	0	...	0	0	0	1	...	0	0	0.5	0.5	...	8	0	0	1	0	...		5	$y^{(4)}$				
$x^{(6)}$	0	0	1	...	1	0	0	0	...	0.5	0	0.5	0	...	9	0	0	0	0	...		1	$y^{(5)}$				
$x^{(7)}$	0	0	1	...	0	0	1	0	...	0.5	0	0.5	0	...	12	1	0	0	0	...		5	$y^{(6)}$				

Fig. 1. Example for sparse real valued feature vectors $\mathbf{x}$ that are created from the transactions of example 1. Every row represents a feature vector $\mathbf{x}^{(i)}$ with its corresponding target $y^{(i)}$. The first 4 columns (blue) represent indicator variables for the active user; the next 5 (red) indicator variables for the active item. The next 5 columns (yellow) hold additional implicit indicators (i.e. other movies the user has rated). One feature (green) represents the time in months. The last 5 columns (brown) have indicators for the last movie the user has rated before the active one. The rightmost column is the target – here the rating.

S. Rendle, "Factorization Machines," 2010 IEEE International Conference on Data Mining, Sydney, NSW, Australia, 2010, pp. 995-1000, doi: 10.1109/ICDM.2010.127. keywords: Mathematical model;Support vector machines;Frequency modulation;Predictive models;Data models;Equations;Computational modeling;factorization machine;sparse data;tensor factorization;support vector machine,

FM

1) Model Equation: The model equation for a factorization machine of degree $d = 2$ is defined as:

$$\hat{y}(\mathbf{x}) := w_0 + \sum_{i=1}^{n} w_i \, x_i + \sum_{i=1}^{n} \sum_{j=i+1}^{n} \langle \mathbf{v}_i, \mathbf{v}_j \rangle \, x_i \, x_j \qquad (1)$$

where the model parameters that have to be estimated are:

$$w_0 \in \mathbb{R}, \quad \mathbf{w} \in \mathbb{R}^n, \quad \mathbf{V} \in \mathbb{R}^{n \times k} \qquad (2)$$

And $\langle \cdot, \cdot \rangle$ is the dot product of two vectors of size k:

$$\langle \mathbf{v}_i, \mathbf{v}_j \rangle := \sum_{f=1}^{k} v_{i,f} \cdot v_{j,f} \qquad (3)$$

A row $\mathbf{v}_i$ within $\mathbf{V}$ describes the i-th variable with k factors. $k \in \mathbb{N}_0^+$ is a hyperparameter that defines the dimensionality of the factorization.

A 2-way FM (degree $d = 2$) captures all single and pairwise interactions between variables:

- w_0 is the global bias.
- w_i models the strength of the i-th variable.
- $\hat{w}_{i,j} := \langle \mathbf{v}_i, \mathbf{v}_j \rangle$ models the interaction between the i-th and j-th variable. Instead of using an own model parameter $w_{i,j} \in \mathbb{R}$ for each interaction, the FM models the interaction by factorizing it. We will see later on, that this is the key point which allows high quality parameter estimates of higher-order interactions ($d \geq 2$) under sparsity.

Modeling the Matrix of Interactions

LightFM

Motivation:

- differences between CF & CBF
- cold-start problem

Solution:

- use both kinds of data (interactions and metadata)
- sum up feature embeddings

Data used:

- item tags
- bag-of-words on 'About Me' section for users

LightFM Features & Embeddings

$$q_u = \sum_{j \in f_u} e_j^U, \quad b_u = \sum_{j \in f_u} b_j^U$$

$$p_i = \sum_{j \in f_i} e_j^I, \quad b_i = \sum_{j \in f_i} b_j^I$$

$$\hat{r}_{ui} = \sigma(q_u \cdot p_i + b_u + b_i)$$

$$\mathcal{L}(e^U, e^I, b^U, b^I) = \prod_{(ui) \in positive} \hat{r}_{ui} \prod_{(ui) \in negative} (1 - \hat{r}_{ui})$$

Kula, Maciej. "Metadata embeddings for user and item cold-start recommendations." arXiv preprint arXiv:1507.08439 (2015).

LightFM jako przykład fuzji informacji

- **Early Fusion**
- Late Fusion

NCF

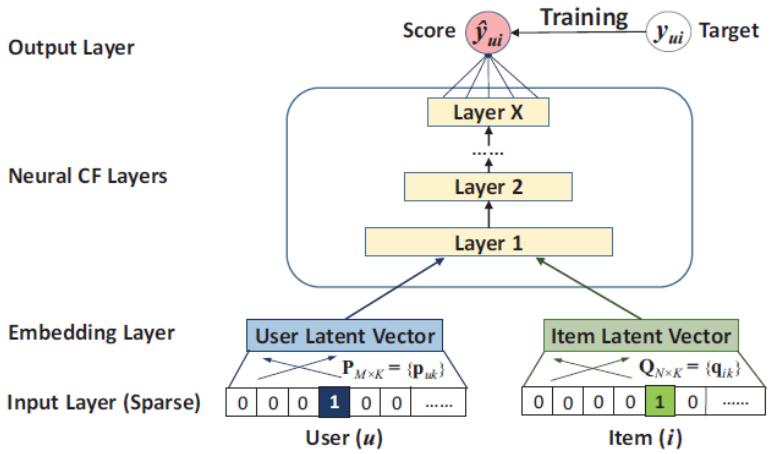


Figure 2: Neural collaborative filtering framework

Xiangnan He, Lizi Liao, Hanwang Zhang, Liqiang Nie, Xia Hu, and Tat-Seng Chua. 2017. Neural Collaborative Filtering. In Proceedings of the 26th International Conference on World Wide Web (WWW '17). International World Wide Web Conferences Steering Committee, Republic and Canton of Geneva, CHE, 173–182. https://doi.org/10.1145/3038912.3052569

NCF Optimization

The loss function is binary cross-entropy:

$$\mathcal{L} = \sum_{(ui) \in positive} y_{ui} \log(\hat{y}_{ui}) + \sum_{(ui) \in negative} (1 - y_{ui}) \log(1 - \hat{y}_{ui})$$

Note! The authors proposed a different approach to combining user and item embeddings. Interestingly, the use of deep learning has become SOTA for RS, however, scoring is still mostly based on the dot product.

NeuMF

Generalized Matrix Factorization – single layer MLP on dot product:

$$\hat{y}_{ui} = \sigma(h^\top (u \circ v))$$

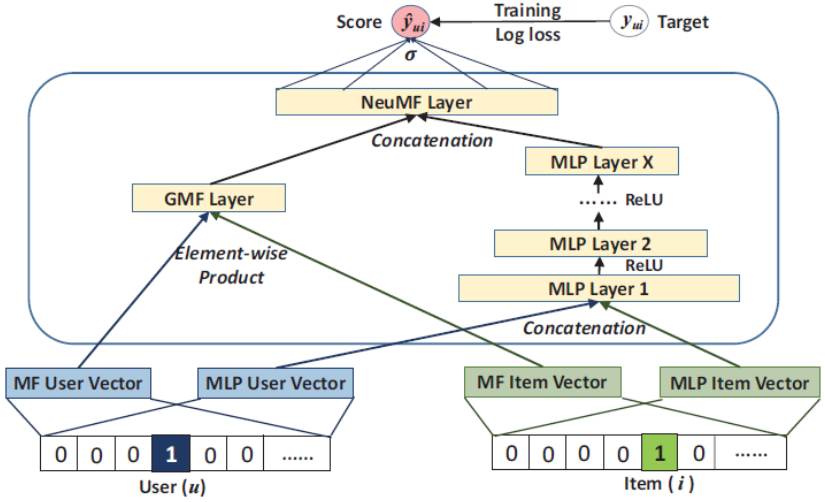


Figure 3: Neural matrix factorization model

MLP: *Tower structure* of layers (half the size in each layer) + ReLU activation.

Note! Both parts are pre-trained separately.

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